

Calibers in GroupLab

The caliber is entered as a cartridge name or as the bullet's diameter. Generated by grouplab calibers from src/GroupLab.Core/Marking/Caliber.cs; do not edit it by hand, because a test fails when this file differs from what the command writes.

1. How to enter a caliber

Type the cartridge's name, such as 6.5 Creedmoor, 308 or 9mm, or the bullet's diameter in inches, such as .264 or 0.264, or in millimeters with mm, such as 6.71 mm. ****A name is matched before a number**** (entry 163 section 3): 6.5 is the 6.5 mm family, 0.264 in, and not 6.5 mm, and .38 is 0.357 in, and not 0.380. As you type, GroupLab lists the cartridges it knows with their diameter beside them, the common first, and under the first it names the family you could pick by mistake.

Only what matches no name is read as a diameter. A designation that names no family in the table is refused rather than read as a diameter, because it is a name written as a number: 7.62 mm could be three different bullets. The designations refused when they name no family are, in inches, .17, .20, .22, .25, .27, .28, .30, .303, .32, .35, .38, .44 and .45, and in millimeters, 5.45 mm, 5.56 mm, 6 mm, 6.5 mm, 6.8 mm, 7 mm, 7.5 mm, 7.62 mm, 7.65 mm, 8 mm, 9 mm and 10 mm. A bare number of one or more that names nothing, such as 7.62 or 22, is refused for the same reason. Every diameter must lie between 0.1 and 1 in.

A designation's refusal names the problem, for example: "7.62 mm is a caliber's name, not the bullet's diameter. Enter the bullet's diameter, such as 7.82 mm or 0.308."

A refusal says what to type: "Enter the bullet diameter in inches, such as 0.308, or in millimeters with mm, such as 7.82 mm."

2. The cartridges GroupLab knows by name

Generated from src/GroupLab.Core/Marking/cartridges.json. Every cartridge here is confirmed at its diameter by both of these sources, and a cartridge only one confirms is held in section 3 until a second does:

- ****A****: Wikipedia, Table of handgun and rifle cartridges, bullet diameter column, which cites SAAMI and CIP. Read 2026-09-24.

- ****B****: Nosler load data, caliber and cartridge index, which lists each cartridge under its bullet diameter from the bullet maker's own load development. Read 2026-09-24.

Bullet	Family	Cartridges	Easily mistaken for
0.204 in (5.18 mm)	.20 calibre	.204 Ruger	
0.224 in (5.69 mm)	.22 centrefire	.223 Remington, .22-250 Remington, .222 Remington, .22 Hornet	
0.243 in (6.17 mm)	6 mm	.243 Winchester, 6mm BR	
0.257 in (6.53 mm)	.25 calibre	.25-06 Remington, .257 Roberts, .257 Weatherby Magnum, .250 Savage	0.264 in
0.264 in (6.71 mm)	6.5 mm	6.5 Creedmoor, 6.5x55 Swedish, .260 Remington, 6.5 Grendel, 26 Nosler	0.257 in
0.277 in (7.04 mm)	.270 calibre	.270 Winchester, 6.8 SPC	0.284 in

0.284 in (7.21 mm)	7 mm	7mm Remington Magnum, 7mm-08 Remington, .280 Remington, 7x57 Mauser	0.277 in
0.308 in (7.82 mm)	.30 calibre	.308 Winchester, .30-06 Springfield, .300 Winchester Magnum, .300 AAC Blackout, .300 WSM, .30-30 Winchester, .300 H&H Magnum	
0.323 in (8.20 mm)	8 mm	8mm Mauser	
0.338 in (8.59 mm)	.338 calibre	.338 Lapua Magnum, .338 Winchester Magnum	
0.355 in (9.02 mm)	9 mm	9mm Luger, .380 ACP, .357 SIG	0.357 in
0.357 in (9.07 mm)	.38 calibre	.38 Special, .357 Magnum	0.355 in
0.375 in (9.53 mm)	.375 calibre	.375 H&H Magnum	
0.400 in (10.16 mm)	.40 calibre	.40 S&W, 10mm Auto	

3. Held back until a second source agrees

Typing one of these works as it did before names were read: GroupLab asks for the diameter. Where the two sources disagree, that is why the cartridge is here, and it is said.

Cartridge	Diameter held	Sources	Why
.17 Remington	0.172 in	B	one source
6.5 PRC	0.264 in	B	one source
6.5x47 Lapua	0.264 in	A	one source
6mm Creedmoor	0.243 in	B	one source
6mm ARC	0.243 in	A	one source
5.56x45mm NATO	0.224 in	A	one source
.22 WMR	0.224 in	A	one source
5.7x28mm	0.224 in	A	one source
22 Nosler	0.224 in	B	one source
.22 Long Rifle	0.222 in	A	A gives 0.223; entry 153 section 4 took 0.222 as nominal. Neither is a second source.
.22 Short	0.222 in	A	one source
7mm PRC	0.284 in	B	one source
.284 Winchester	0.284 in	B	one source
28 Nosler	0.284 in	B	one source
.300 PRC	0.308 in	B	one source
.300 Norma Magnum	0.308 in	A	one source
.30 Carbine	0.308 in	A	one source
7.62x39mm	0.310 in	A, B	The two disagree: A gives 0.312, B lists it under 0.310. Held until a third source settles it.
7.62x54mmR	0.311 in	A	A gives 0.308, which is not what anybody loads it with; B does not list it. Held.
.303 British	0.311 in	A	one source
.338 Norma Magnum	0.338 in	A	one source

.44 Magnum	0.429 in	B	one source
.44 Special	0.429 in	B	one source
.45 ACP	0.451 in	B	one source
.45 Colt	0.451 in	B	B lists 0.451; 0.452 and 0.454 are also loaded.
.17 HMR	0.172 in	A	one source
.50 BMG	0.510 in	none yet	one source
6.8 Western	0.277 in	none yet	one source
.25 Creedmoor	0.257 in	none yet	one source
22 ARC	0.224 in	none yet	one source
.224 Valkyrie	0.224 in	none yet	one source
8.6 Blackout	0.338 in	none yet	one source
6 Dasher	0.243 in	none yet	one source

4. The diameters in the pick list

Every distinct bullet diameter in Alan's rifle and pistol lists, smallest first. Any other diameter in range can be typed.

Inches	Millimeters
.172	4.37
.204	5.18
.2215	5.63
.222	5.64
.224	5.69
.243	6.17
.257	6.53
.264	6.71
.277	7.04
.284	7.21
.308	7.82
.309	7.85
.310	7.87
.3105	7.89
.312	7.92
.321	8.15
.323	8.20
.338	8.59
.355	9.02
.356	9.04
.357	9.07
.358	9.09

.366	9.30
.375	9.53
.400	10.16
.410	10.41
.411	10.44
.416	10.57
.423	10.74
.430	10.92
.451	11.46
.452	11.48
.454	11.53
.458	11.63
.474	12.04
.500	12.70
.505	12.83
.510	12.95

5. Why names came back

Alan decided in entry 107 that the caliber is a diameter and nothing else, reversing entries 105 and 106, which read names through a table whose defects all came from reading a name as a diameter. Entry 108 refused the designations a person might type as a diameter. **Entry 163 reverses both, on the evidence of the first real user**: a person who shoots a 6.5 Creedmoor typed 6.5 and was offered .257, because the pick list matched 6.53 mm. A name he knows is a better thing to ask for than a diameter he does not, provided the table is right, which is why every name in it has two sources and every disagreement between them keeps the name out.